Lab Work Assignment 5: Weights and Biases Experiment Tracking and Artifact Management

Course: ML Operations

Objective:
The objective of this lab is to set up Weights & Biases (W&B) for experiment tracking, log relevant artifacts (such as configurations, parameters, metrics, and files), and demonstrate effective experiment management. Students are expected to integrate W&B into their machine learning workflows to enhance reproducibility, track experiments, and log essential information for model training and evaluation.

Resources:

- [W&B Documentation]
- [Example W&B Project]

Submission Deadline: 04 November

Part 1: Weights & Biases Setup

Tracking Setup

Task:

- Install Weights & Biases in your environment.
- Set up a W&B account and create a new project in W&B.
- Ensure you correctly initialize W&B in your code to log experiment data.
- Demonstrate that your setup is correctly integrated by showing that the W&B dashboard displays your logs and runs.

Version Control Setup

Task:

- Use GitHub (or a similar platform) to create a repository for your project.
- Add necessary code and configuration files to the repository, and ensure that you exclude unnecessary files (e.g., large datasets, temporary files) using a `.gitignore` file.

Part 2: Logging Parameters, Configs, and Metrics in W&B

Logging Parameters and Configurations

Task:

- Create a configuration file (e.g., `config.yaml` or `config.json`) to manage important model and training parameters, such as learning rates, batch sizes, and data paths.
- Log these parameters using W&B's `wandb.config` object for each experiment run.

Metrics Logging

Task:

- Log key metrics (e.g., accuracy, loss, or any relevant evaluation scores) to W&B.
- Define an appropriate interval for logging metrics (e.g., per epoch or after every n iterations). Ensure these metrics are visualized in the W&B dashboard.
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Part 3: Logging Output Artifacts

Model Artifact Logging

Task:

- Ensure that your best-trained model (e.g., weights and architecture) is logged as an artifact in W&B.
- Demonstrate that the logged model can be downloaded from W&B and used for further evaluation or deployment.

Visualizing Artifacts (Optional)

Task (Optional, for additional points):

- Create custom visualizations of artifacts, such as performance metrics or comparison graphs of different runs, using the W&B dashboard.

Part 4: Experimentation with W&B

Experiment Creation and Run Management

Task:

- Log multiple runs under the same W&B project. Each run should correspond to different configurations or parameter values, demonstrating experimentation and tuning.
- Compare the logged runs in the W&B dashboard to analyze model performance.

Run Naming (Additional Points)

Task (Optional, for additional points):

- Use a structured naming convention for your runs (e.g., “Run 1 - Default Config”, “Run 2 - Tuned Config”).
- Demonstrate how consistent run names improve experiment tracking and comparison within the W&B UI.

Part 5: Lab Report

Content:

- **Introduction**: Explain the importance of experiment tracking in machine learning workflows and introduce Weights & Biases as a tool for this purpose.
- **Tracking Setup**: Describe how you set up W&B and integrated it into your workflow.
- **Logging Details**: Provide an overview of the parameters, metrics, and artifacts you logged during the experiment.
- **Experimentation Process**: Discuss how you managed multiple runs within the same project and how the logged information helped with model evaluation.
- **Reflection**: Reflect on the benefits and challenges of using W&B in your project, including any additional improvements that could be made.

Grading Criteria

- **Weights & Biases Setup**: Proper setup and demonstration of W&B integration in your project.
- **Experiment Logging**: Successful logging of parameters, metrics, and artifacts, and appropriate use of the W&B UI to track and compare runs.
- **Version Control**: Effective use of GitHub (or similar) for code management, including clear commit messages and an organized repository.
- **Lab Report**: Clarity, depth, and completeness of the lab report, including reflection on the benefits and challenges of using W&B.